

Dalam kategori Crypto pada **CTF Jeopardy**, **Classical Ciphers** adalah teknik kriptografi klasik sebelum era modern AES/RSA.

Biasanya challenge meminta peserta:

- mengenali jenis cipher,
- menemukan key,
- melakukan decrypt,
- atau brute force.

Cipher klasik sering muncul di:

- Beginner Crypto
- Warmup challenge
- Puzzle challenge

Apa Itu Classical Cipher?

Cipher klasik adalah metode menyamarkan teks dengan:

- menggeser huruf,
- mengganti huruf,
- menggunakan tabel alfabet,
- atau operasi sederhana.

Contoh:

HELLO

↓ Caesar +3

KHOOR

1. Caesar Cipher

Cipher paling dasar.

Cara Kerja

Setiap huruf digeser sejumlah tertentu.

Contoh shift +3:

A → D

B → E

C → F

Contoh

Plaintext:

FLAG

Encrypt +3:

IODJ

Cara Decrypt Caesar

Brute Force

Karena hanya ada 26 shift.

Python

```
cipher = "IODJ"
```

```
for shift in range(26):
```

```
    result = ""
```

```
    for c in cipher:
```

```
        result += chr((ord(c)-65-shift)%26 + 65)
```

```
    print(shift, result)
```

CTF Example

Soal

SYNT{pnrfne_pvcure}

Hint:

ROT13

ROT13 adalah Caesar shift 13.

Flag:

FLAG{caesar_cipher}

2. Atbash Cipher

Cipher membalik alfabet.

A $\leftrightarrow$ Z

B $\leftrightarrow$ Y

C $\leftrightarrow$ X

Contoh

Plaintext:

FLAG

Encrypt:

UOZT

Python Atbash

```
import string
```

```
alphabet = string.ascii_uppercase
```

```
reverse = alphabet[::-1]
```

```
table = str.maketrans(alphabet, reverse)
```

```
cipher = "UOZT"
```

```
print(cipher.translate(table))
```

Ciri Atbash

- tidak butuh key,
- pola sering aneh,
- frekuensi huruf tetap.

3. Vigenère Cipher

Versi lebih kompleks dari Caesar.

Menggunakan:

- plaintext,
- keyword.

Rumus

Cipher = Plaintext + Key

Contoh

Plaintext:

HELLO

Key:

KEYKE

Encrypt:

RIJVS

Tabel Vigenère

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Message Character

Key Character

Using the Table

Encryption

Message: S E N D H E L P

Key: B U L G E B U L

Ciphertext: T Y Y J L F F A

Decryption

Ciphertext: T Y Y J L F F A

Key: B U L G E B U L

Message: S E N D H E L P

Plain-text Letter																											
Key Letter	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	
	B	A	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	
	C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	
	D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	
	E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	
	F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	
	G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	
	H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	
	I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	
	J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	
	K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	
	L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	
	M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	
	N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	
	O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	
	P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	
	Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	
	R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	
	S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	
	T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	
	U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	
	V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	
	W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	
	X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T			

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y


```
def vigenere_decrypt(cipher, key):  
    result = ""  
  
    key = key.upper()  
    key_index = 0  
  
    for c in cipher:  
        shift = ord(key[key_index % len(key)]) - 65  
        plain = chr((ord(c)-65-shift)%26 + 65)  
  
        result += plain  
        key_index += 1  
  
    return result  
  
print(vigenere_decrypt("RIJVS","KEY"))
```

Teknik Solve Vigenère di CTF

Jika key diketahui

Langsung decrypt.

Jika key tidak diketahui

Gunakan:

- Kasiski Examination,
 - Index of Coincidence,
 - frequency analysis.
-

Tools

- CyberChef
- dCode

[dCode Vigenere Cipher Tool](#)

4. Affine Cipher

Menggunakan rumus matematika.

Formula Encrypt

$$E(x) = (ax + b) \bmod 26$$

Formula Decrypt

$$D(x) = a^{-1}(x - b) \bmod 26$$

Contoh

Key:

$$a = 5$$

$$b = 8$$

Huruf:

$$A = 0$$

$$B = 1$$

...

Ciri Affine

- ada operasi linear,
 - mirip Caesar tetapi lebih kompleks.
-

Python Affine

```
def affine_encrypt(text, a, b):
```

```
result = ""
```

```
for c in text:
```

```
    x = ord(c)-65
```

```
    result += chr((a*x+b)%26 + 65)
```

```
return result
```

```
print(affine_encrypt("FLAG",5,8))
```

5. Substitution Cipher

Setiap huruf diganti huruf lain.

Contoh

A → Q

B → W

C → E

Plaintext:

HELLO

Cipher:

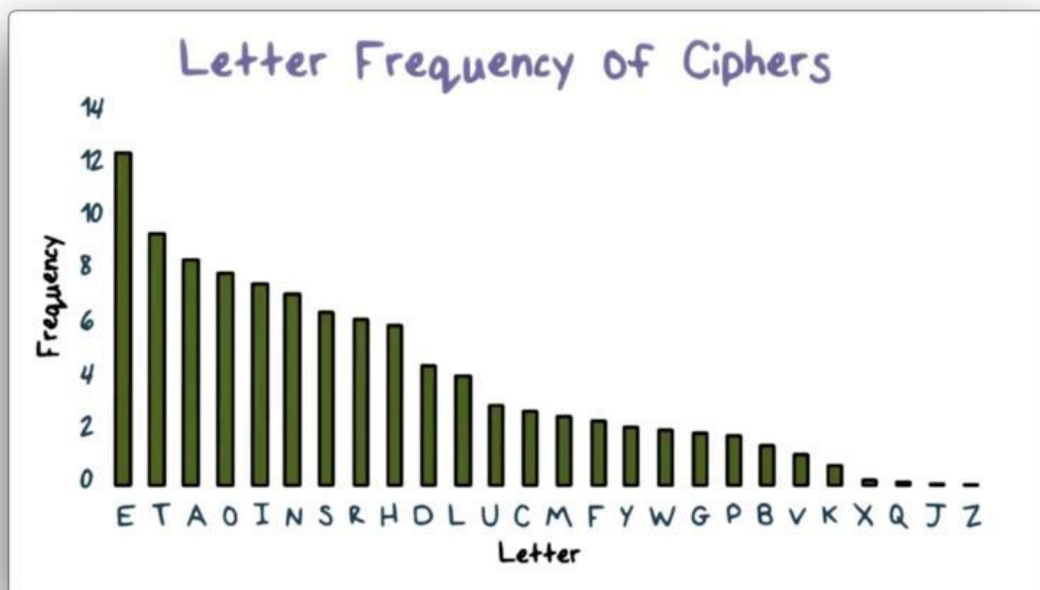
ITSSG

Teknik Solve

Biasanya memakai:

- frequency analysis,
- tebakan kata umum,
- pola flag.

Frequency Analysis

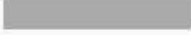
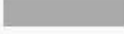
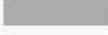
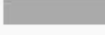


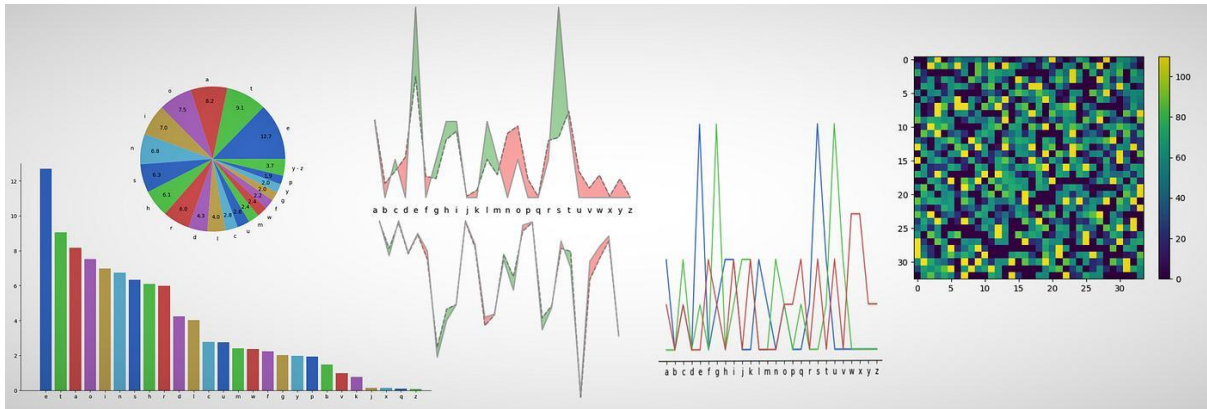
Frequency Analysis

- Substitution cipher is vulnerable to **frequency analysis**
- Note that in the hello_world example below, symbol “o” appears 2 times in the plaintext, whereas the corresponding “q” also appears 2 times in the ciphertext

plaintext: h e l l o _ w o r l d
ciphertext: h n l l q p o q i l b

- **A monoalphabetic cipher:** the substitution is fixed for *each letter* of the alphabet
- Suppose the plaintexts are English sentences. The **letter frequency distribution** in English text is **not** uniform, for e.g. “e” is more commonly used than “z”.

Letter ↕	Relative frequency in the English language ▼	
e	12.702%	
t	9.056%	
a	8.167%	
o	7.507%	
i	6.966%	
n	6.749%	
s	6.327%	
h	6.094%	
r	5.987%	
d	4.253%	
l	4.025%	
c	2.782%	
u	2.758%	
m	2.406%	
w	2.361%	
f	2.228%	
g	2.015%	
y	1.974%	
p	1.929%	
b	1.492%	
v	0.978%	
k	0.772%	
j	0.153%	
x	0.150%	
q	0.095%	
z	0.074%	



Huruf paling umum bahasa Inggris:

E T A O I N

Tools

[quipqiup Substitution Solver](#)

6. XOR Cipher

Sangat sering muncul di CTF modern.

Konsep XOR

$A \oplus B = C$

XOR Truth Table

A B XOR

0 0 0

0 1 1

1 0 1

1 1 0

Contoh XOR

A = 65

Key = 1

65 XOR 1 = 64

Python XOR

```
cipher = [71,77,64,70]
```

```
key = 1
```

```
plain = ''.join(chr(c ^ key) for c in cipher)
```

```
print(plain)
```

XOR di CTF

Biasanya:

- single-byte XOR,
 - repeating XOR,
 - XOR + Base64,
 - XOR + Hex.
-

Teknik Solve XOR

Single Byte XOR

Bruteforce:

```
for key in range(256):
```

```
...
```

Repeating XOR

Gunakan:

- Hamming distance,

- frequency analysis.

Challenge XOR Contoh

Cipher

0d0c0a091e

Hint:

single-byte XOR

Solve

1. Convert hex
 2. XOR semua kemungkinan
 3. Cari output readable
-

Cara Identifikasi Cipher di CTF

Cipher	Ciri
Caesar	huruf masih readable
ROT13	setengah readable
Atbash	alfabet terbalik
Vigenère	mirip random
XOR	sering hex/binary
Substitution	pola huruf tetap

Workflow Solve Crypto CTF

1. Identifikasi Format

- hex?
- base64?
- pure text?

2. Cari Pola

- karakter berulang,
- panjang,
- readable text.

3. Gunakan Tools

CyberChef

[CyberChef](#)

dCode

[dCode Classical Ciphers](#)

Contoh Challenge Mini

Soal

Uifsf jt b tfdsfu gmbh

Analisis

Huruf bergeser +1.

Decrypt

There is a secret flag

Kesalahan Umum Pemula

Mengira semua random text adalah Base64

Padahal bisa:

- XOR,
- Vigenère,
- Substitution.

Tidak mencoba brute force Caesar

Padahal cepat.

Tidak melakukan frequency analysis

Padahal sangat penting.

Kesimpulan

Classical Cipher yang paling sering muncul di CTF:

1. Caesar
2. ROT13
3. Atbash
4. Vigenère
5. Affine
6. Substitution
7. XOR

Skill penting:

- pattern recognition,
- brute force,
- frequency analysis,
- penggunaan CyberChef/dCode,
- scripting Python.

Kalau mau, saya juga bisa bantu:

- membuat challenge crypto sendiri,
- walkthrough challenge crypto nyata,
- latihan soal Caesar/Vigenère/XOR,
- atau membuat toolkit Python otomatis untuk solve cipher CTF.